



Bearing Selection & Clearance Reference

Quick-reference guide for journal bearing selection, radial clearances, and axial (thrust) float across common steam turbine shaft diameters and OEM families.

Basis: figures below follow the general industry rule of thumb for fluid-film (babbitt) journal bearings in process machinery — a diametral clearance of roughly 1.0 to 1.5 mils per inch of shaft diameter (0.001 to 0.0015 x diameter), tightening toward the lower end for slower, heavier-loaded shafts and opening toward the upper end for higher-speed shafts. Rotor balance quality generally follows ISO 21940-11 Grade G2.5 for turbine rotors. These are widely used starting points, not a substitute for the OEM's own clearance chart.

1. Journal Bearing Radial Clearance by Shaft Diameter

Shaft Dia. (mm)	Bearing Type	Radial Clearance (mm)	Typical Application
25 - 50	Sleeve / plain babbitt	0.025 - 0.05	Small aux. turbines, pumps
50 - 100	Sleeve / 2-lobe	0.05 - 0.09	5-15 MW turbine journals
100 - 150	Elliptical / tilting pad	0.09 - 0.14	15-30 MW turbine journals
150 - 250	Tilting pad	0.14 - 0.22	30-60 MW turbine journals
250 - 400	Tilting pad (multi-pad)	0.22 - 0.32	Large TG sets, generator end

General guide only — clearance is normally 0.10-0.15% of shaft diameter for plain sleeve bearings and slightly tighter for tilting-pad designs. Always confirm against the OEM's clearance chart for the specific turbine model.

1a. Worked Example: Clearance Formula

Diametral clearance (mm) = shaft diameter (mm) x 0.001 to 0.0015. For a 150 mm journal: 150 x 0.001 = 0.15 mm (tight end) to 150 x 0.0015 = 0.225 mm (open end) — matching the 0.09-0.14 mm range shown for elliptical/tilting-pad bearings in that diameter band above once the bearing-type adjustment is applied. Remember this is the **cold, as-assembled** clearance: the running (hot) clearance will be smaller once the shaft and bearing reach operating temperature, since the shaft normally grows faster than the housing. Always re-check the OEM's distinction between assembly clearance and operating clearance before rejecting a bearing on a cold measurement alone.

2. Thrust Bearing Axial Float

Turbine Rating	Thrust Bearing Type	Axial Float (mm)
Up to 5 MW	Tapered land / fixed pad	0.25 - 0.35
5 - 20 MW	Tilting pad (Kingsbury type)	0.30 - 0.45
20 - 60 MW	Tilting pad, self-equalizing	0.35 - 0.55

3. Babbitt Wear & Rejection Limits

Symptom	Likely Cause	Action
Wire-drawing / streaking	Electrical discharge machining (shaft currents)	Check earthing brush, insulate bearing



Symptom	Likely Cause	Action
Wiping (smeared babbitt)	Oil starvation, overheating, misalignment	Re-babbitt, check oil supply & alignment
Fatigue cracking	Cyclic loading beyond fatigue limit	Re-babbitt, review vibration levels
Pitting / corrosion	Water contamination, chemical attack	Flush oil system, verify moisture content

4. Rebabbiting & Bearing Services Offered by Keshav Enterprises

Service	Scope
Rebabbiting	Strip, re-tin, re-pour tin/lead babbitt on shells and pads to OEM ratio
Journal polishing	In-situ and shop shaft journal polishing to remove scoring
Clearance measurement	Plastigauge / lead-wire method with certified report
Bearing manufacture	Reverse-engineered sleeve, elliptical, and tilting-pad bearings

5. Related Rotor Balance Quality Grades (ISO 21940-11)

Bearing wear accelerates sharply on an out-of-balance rotor, so re-balancing after any bearing work is good practice. ISO 21940-11 (formerly ISO 1940-1) sets the permissible residual unbalance as $U_{per} (g \cdot mm) = 9549 \times G \times M / n$, where G is the grade in mm/s, M is rotor mass in kg, and n is maximum service speed in RPM.

Rotor / Machine Type	Typical ISO 21940-11 Grade
Steam and gas turbine rotors, turbocompressors	G2.5
General industrial rotors (fans, pump impellers, motors)	G6.3
Grinding machine drives, precision spindles	G1.0 or finer
Crankshafts, marine diesel drives (rigid, low-speed)	G16 to G40

Figures in this reference are typical industry ranges for common steam turbine journal and thrust bearings. They are not a substitute for OEM documentation — always verify against the specific turbine manufacturer's clearance chart before machining, shimming, or rebabbiting. Contact Keshav Enterprises for a model-specific clearance chart or reverse-engineering support.